ASSEMBLY REQUEST

The prompt

This report was produced by Claude, working through the PACER 3.2.0 automated test assembly tool, in response to the request below, grouped into the steps it asks for.

GOAL	I want to build test forms using PACER's ATA functions.
ITEM BANK	I have an item bank here `C:\PACER\IrtEstimation\data\ATA\bank_pvalue.csv`
BLUEPRINT	and I have a test blueprint I want to satisfy here `C:\PACER\IrtEstimation\data\ATA\blueprint_pvalue.csv`.
TIF TARGETS	In addition, I want the test information function to meet specific statistical criteria at various points on the ability scale. Those details are in this file `C:\PACER\IrtEstimation\data\ATA\tif_spec.csv`.
FORMS & OVERLAP	I would like for you to build three (3) test forms. You may use between 10 to 15 common items between them.
SOLVER	Let the solver run for up to 3 minutes to find the best solution.
REPORT	Generate a complete report as a PDF with tables that show the items that were chosen for each form and show which items are common across all forms. Show if the forms meet blueprints and the TIF specifications.
PERSPECTIVE	Share with me your perspectives on this task and if there are concerns or issues that make building these forms difficult.

Three linked forms from a 100-item bank

Three 20-item forms assembled simultaneously with PACER's CP-SAT solver against a four-strand blueprint, a p-value sum constraint, a seven-point TIF band, and 10–15 linking items between forms.

100 items + 1 stimulus group **3** forms × **20** items Objective: maximize TIF at θ = 0

Solver: Optimal in 44 s (limit 180 s)

Bottom line

1. **All three forms satisfy every constraint.** The solver proved the solution optimal in 44 seconds. Each form meets the strand, item-type, p-value, overlap, and all seven TIF requirements, and peak information is 5.88–5.90 at θ ≈ 0 on every form.
2. **The forms are near-copies of each other.** Each pair of forms shares 15 items (the maximum allowed), 14 items are on all three forms, and each form has only 4 items of its own. Only 29 distinct items are used across the 60 slots.
3. **The p-value column doesn't agree with the IRT parameters.** Bank p-values correlate 0.01 with the IRT-implied proportion correct. The p-value constraint is binding, and it pulls in items such as 60 and 65 (p = .99) whose IRT parameters say they are of medium or hard difficulty. This needs to be resolved before the p-value rule means anything.
4. **The bank is thin where the blueprint needs it most.** Every form uses the same five polytomous items (40% of score points), the TIF minimum at θ = −2 is met by only 0.02–0.11, and half the bank contributes almost no information at θ = 0.

18 / 18	5.88	14	r = .01
constraint checks passed per form	TIF at θ = 0, lowest of the three forms	items common to all three forms	bank p-value vs. IRT-implied p

1 Inputs

Item bank

The bank has 100 scorable items: 92 dichotomous 3PL items (1 point each) and 8 polytomous GPCM items (2 points each), split evenly across four strands. It also contains one stimulus group (G1, items 6, 7, 8, assembled all-or-nothing) and one enemy set (items 4, 5, and 99, at most one per form).

Strand	3PL items	GPCM items	Total	Median a	b range (3PL)	Mean p-value
Strand 1	23	2	25	0.65	-3.76 to 3.93	0.56
Strand 2	23	2	25	0.55	-3.47 to 3.39	0.58
Strand 3	23	2	25	0.54	-3.94 to 3.27	0.48
Strand 4	23	2	25	0.52	-3.85 to 3.81	0.48
Bank	**92**	**8**	**100**	**0.55**		**0.52**

Blueprint

Constraint	Required per form	What it implies
Total items	exactly 20	Fixed length
Strand 1, 2, 3, 4	exactly 5 each	Balanced content
Dichotomous items	0–15	Together with the 20-item total, these force exactly 15 dichotomous and 5 polytomous items
Polytomous items	0–5	
Sum of p-values	12–14	Mean p of .60–.70; the bank mean is .52
Objective	maximize TIF at θ = 0	Most information at the center of the scale

TIF specification

θ	−3	−2	−1	0	1	2	3
Minimum information	0	3	2	2	2	2	0
Maximum information	8	12	10	10	10	12	8
Label	low range	below average	lower target	target region	upper target	above average	high range

The minimum at θ = −2 (3.0) is higher than at θ = −1, 0, or 1 (2.0). That asks for more measurement precision below average than at the center, which is unusual. Section 4 shows it is the hardest TIF requirement to meet.

Assembly settings

Engine	PACER 3.2.0 `assemble_test` (Google OR-Tools CP-SAT)
Forms	3, assembled simultaneously
Linking	On: 10–15 shared items for each consecutive pair (A–B, B–C); A–C limited to at most 15
Stimulus groups	Strict (all items in G1 or none)
Scaling constant	D = 1.7
Time limit	180 seconds (the solver proved optimality in 44 s)

2 The assembled forms

The table lists every item used on at least one form. Shaded rows are on all three forms.

Item	Strand	Model	Points	a	b	c	p-value	Form A	Form B	Form C
9	Strand 1	3PL	1	0.87	-2.71	0.02	0.98	●		●
10	Strand 1	3PL	1	0.65	0.45	0.00	0.13			●
16	Strand 1	3PL	1	0.79	-0.60	0.08	0.49	●	●	●
17	Strand 1	3PL	1	0.69	-1.57	0.18	0.82		●	
20	Strand 1	3PL	1	0.48	0.45	0.03	0.83	●	●	
22	Strand 1	3PL	1	0.92	-0.74	0.06	0.24	●	●	●
24	Strand 1	3PL	1	0.59	-1.11	0.03	0.92	●	●	●
32	Strand 2	3PL	1	0.99	-0.07	0.07	0.08	●	●	●
34	Strand 2	3PL	1	0.74	-1.61	0.18	0.77		●	●
35	Strand 2	GPCM	2	0.87	d: 1.05, 2.65	—	0.47	●	●	●
36	Strand 2	GPCM	2	0.56	d: -1.09, -0.55	—	0.80	●	●	●
38	Strand 2	3PL	1	0.70	-0.83	0.13	0.70	●	●	●
39	Strand 2	3PL	1	0.63	0.16	0.06	0.18	●		
52	Strand 3	3PL	1	0.57	0.32	0.10	0.42		●	
57	Strand 3	3PL	1	0.77	-0.31	0.01	0.70	●	●	●
60	Strand 3	3PL	1	0.25	-0.14	0.03	0.99	●		
65	Strand 3	3PL	1	0.55	2.51	0.00	0.99			●
66	Strand 3	3PL	1	0.81	0.76	0.20	0.75	●	●	●
68	Strand 3	GPCM	2	0.77	d: 0.34, 1.02	—	0.40	●	●	●
72	Strand 3	3PL	1	0.82	-1.58	0.09	0.62	●	●	●
79	Strand 4	3PL	1	0.57	-0.92	0.07	0.40			●
83	Strand 4	3PL	1	0.94	-1.60	0.12	0.36		●	
84	Strand 4	3PL	1	0.66	0.06	0.17	0.52	●		
87	Strand 4	3PL	1	0.95	0.16	0.11	0.49	●	●	●
95	Strand 4	3PL	1	0.45	-2.17	0.10	0.74			●

Item	Strand	Model	Points	a	b	c	p-value	Form A	Form B	Form C
97	Strand 4	3PL	1	0.86	-2.28	0.03	0.60	●		
98	Strand 4	3PL	1	0.52	-3.85	0.19	0.85		●	
99	Strand 4	GPCM	2	0.46	d: -1.34, 0.90	—	0.71	●	●	●
100	Strand 4	GPCM	2	0.58	d: -1.28, 1.70	—	0.66	●	●	●

Overlap between forms

Shared items	Form A	Form B	Form C
Form A	20	15	15
Form B	15	20	15
Form C	15	15	20

On all three forms	**14**
On exactly two forms	3
Unique to one form	4 + 4 + 4
Distinct items used	29 of 100

Item group	Count	Items
Common to all three forms	14	16, 22, 24, 32, 35, 36, 38, 57, 66, 68, 72, 87, 99, 100
Forms A and B only	1	20
Forms A and C only	1	9
Forms B and C only	1	34
Form A only	4	39, 60, 84, 97
Form B only	4	17, 52, 83, 98
Form C only	4	10, 65, 79, 95

Your request allowed 10 to 15 common items. The solver went to the top of that range on every pair, because the objective rewards reusing the most informative items. The 14 items common to all three forms also fall inside 10–15, so the build satisfies the request under either reading: shared items per pair, or items shared by all three forms.

3 Blueprint compliance

Constraint	Required	Form A	Form B	Form C	Status
Total items	20	20	20	20	✓ Met
Strand 1	5	5	5	5	✓ Met
Strand 2	5	5	5	5	✓ Met
Strand 3	5	5	5	5	✓ Met
Strand 4	5	5	5	5	✓ Met
Dichotomous items	0–15	15	15	15	✓ Met
Polytomous items	0–5	5	5	5	✓ Met
Sum of p-values	12–14	12.14	12.08	12.04	✓ Met
Score points	—	25	25	25	—
Enemy set {4, 5, 99}	≤ 1	1	1	1	✓ Met
Stimulus group G1 (6, 7, 8)	all or none	0	0	0	✓ Met
Shared items A–B / B–C	10–15	15 / 15 (A–C: 15)			✓ Met

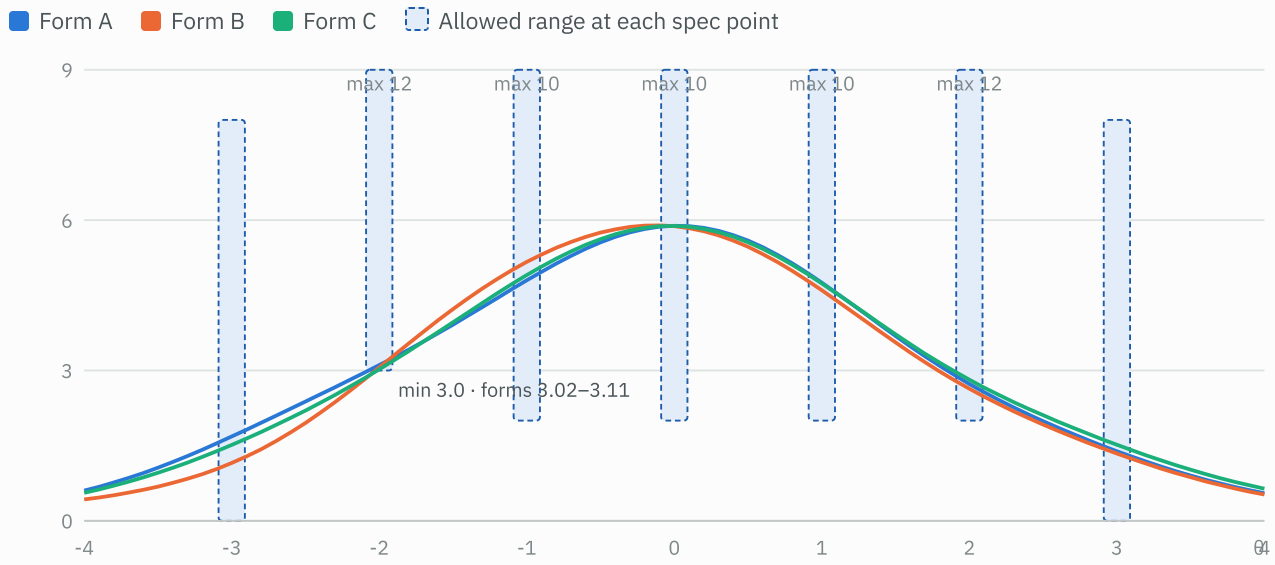
The p-value sums of 12.04–12.14 sit just above the 12.0 floor on every form, so that constraint is binding. When I removed it in a sensitivity run, the solver chose forms with p-value sums of 9.4–10.0 (Section 6). The 15/5 split between dichotomous and polytomous items is also binding, as described in Section 1.

4 TIF compliance

θ	Required	Form A	Form B	Form C	Smallest margin	Status
-3	0–8	1.68	1.15	1.51	1.15	✓ Met
-2	3–12	3.11	3.06	3.02	0.02	▲ Met, tight
-1	2–10	4.80	5.17	4.90	2.80	✓ Met
0	2–10	5.89	5.88	5.89	3.88	✓ Met
1	2–10	4.75	4.60	4.74	2.60	✓ Met
2	2–12	2.73	2.64	2.82	0.64	✓ Met
3	0–8	1.39	1.34	1.52	1.34	✓ Met

Values are exact information at each θ. I recomputed them from the bank's item parameters (3PL and GPCM, D = 1.7), and they match PACER's curve output to four decimals. Smallest margin is the distance from the nearest bound across the three forms.

Figure 1. Test information functions with the TIF specification



All three curves pass through every allowed range. The forms nearly coincide between θ = −1 and θ = 2, which is expected given 14 shared items. The binding point is θ = −2, where every form sits just above the 3.0 minimum. No form comes near any maximum.

Figure 2. Standard error of measurement

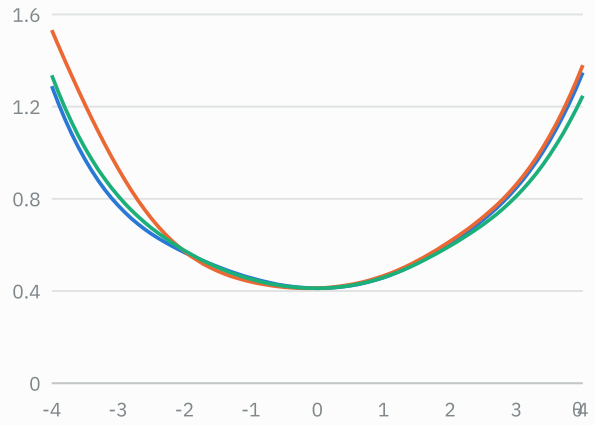
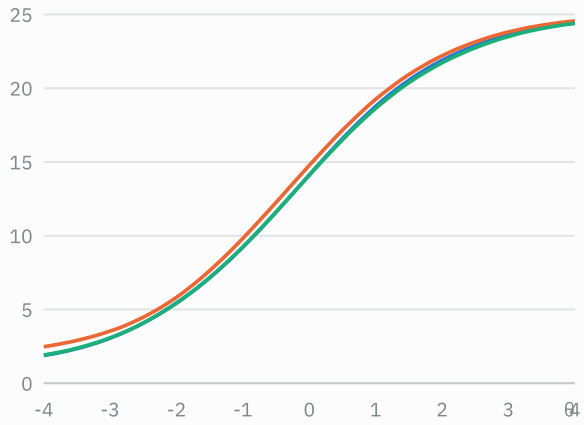


Figure 3. Test characteristic curves (expected raw score, max 25)



SEM is 0.41 at θ = 0 and rises to 0.57–0.62 at θ = ±2. The TCCs are close but not identical: at θ = 0 Form B's expected score is 14.77, versus 14.12 for A and 14.10 for C, a 0.67-point gap. That is the largest gap anywhere on the scale. Raw scores on different forms still need equating before they can be compared.

5 Perspectives and concerns

The build succeeded, but several features of the bank and blueprint made it harder than it looks, and some deserve attention before these forms go into use. They're listed from most to least serious.

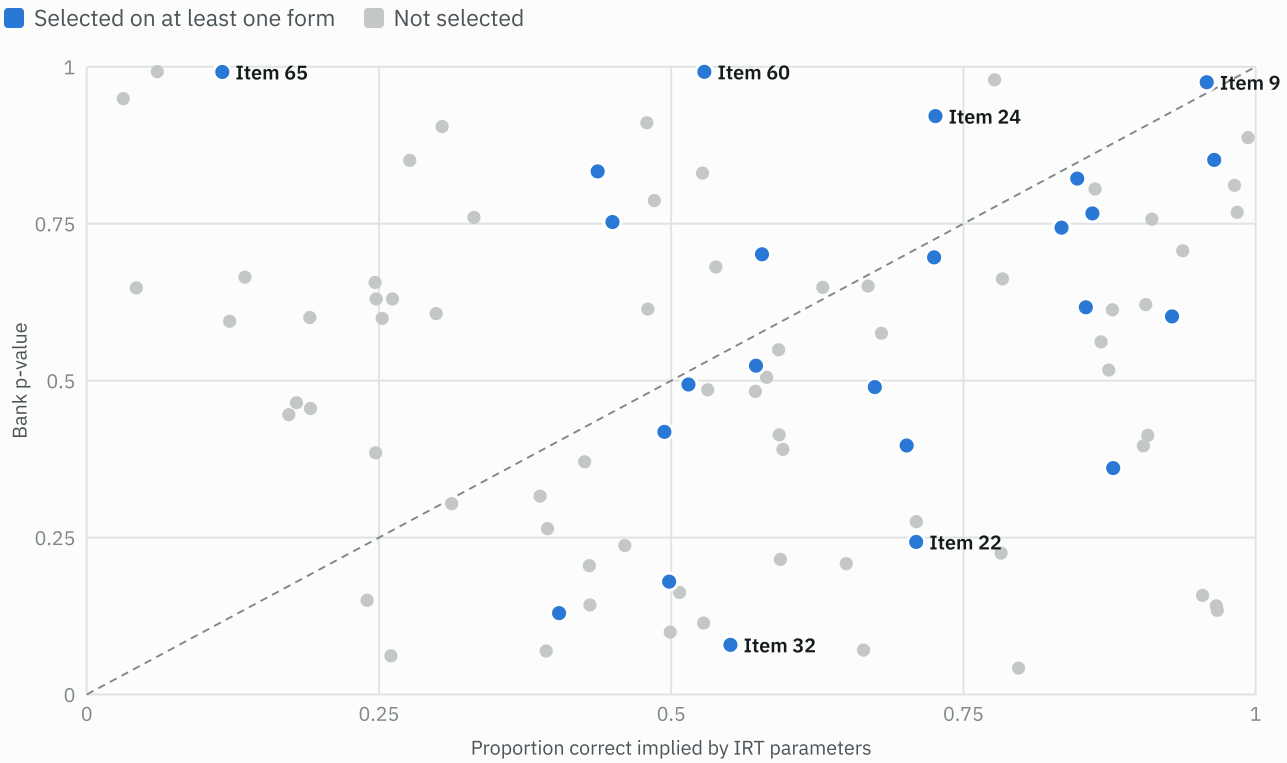
1 **The p-value column is inconsistent with the IRT parameters**

For every 3PL item I computed the proportion correct implied by its IRT parameters for a standard normal population. The correlation between that and the bank's p-value column is **0.01**; the correlation with *b* is −0.03. In 27 of 92 items the two disagree by more than 0.40. Some examples on these forms:

- Item 65: p = .99, but b = 2.51 implies about 12% correct.
- Item 60: p = .99, but b = −0.14 with a = 0.25 implies about 53%.
- Item 32 (on all three forms): p = .08, but b = −0.07 with a = 0.99 implies about 55%.

Because the p-value sum is a binding constraint, the solver satisfies it by picking items with high listed p-values, whether or not they are easy. The constraint currently controls a number that doesn't describe how the items will behave. Either the p-values come from a different population or calibration, or the columns are misaligned. Figure 4 shows the scatter.

Figure 4. Bank p-value vs. proportion correct implied by IRT parameters (3PL items, θ ~ N(0,1))



If the two sources agreed, points would fall near the diagonal. They are scattered across the whole square. Labeled points are discussed in the text.

2 **The forms are close to clones**

With the overlap allowed up to 15, the information objective drives every pair to exactly 15 shared items. Only 4 items per form are unique, and 71 of the 100 bank items were never used. Three forms with this much overlap give little protection against item exposure: anyone who has seen one form has seen 75% of the next. Forcing the minimum overlap of 10 still met every constraint, cut items common to all three forms from 14 to 7, and raised distinct items from 29 to 37. The cost was about 8% of the information at $\theta = 0$ (5.40 vs. 5.88; Section 6).

3 **Polytomous items are too scarce for the blueprint**

The blueprint effectively requires exactly 5 polytomous items per form, and the bank has only 8. Items 4, 5, and 99 are mutually exclusive enemies, and item 67 is barely informative ($a = 0.23$). That leaves at most six usable GPCM items per form. All three forms carry the identical set (35, 36, 68, 99, 100), which accounts for 10 of the 25 score points. Those points are fully exposed across forms and can't vary.

4 **The bank has limited information, especially in the tails**

Discrimination is modest (median $a = 0.54$, maximum about 1.0), and many b-values are extreme. Half the items contribute less than 0.04 information at $\theta = 0$. Even a single form built only for $\theta = -2$, ignoring overlap and item-type rules, could reach at most about 5.1 there. The three-form build reaches 3.02–3.11 against a 3.0 floor. Raising that floor, or adding a fourth form, would probably make the problem infeasible. The upper TIF bounds (8–12) are never close to binding, so they don't constrain the build.

5 **Smaller structural points**

- **The linking rule covers consecutive pairs only.** PACER enforces the 10–15 overlap on A–B and B–C, and caps A–C at 15 with no minimum. A–C landed at 15 here, but in other runs it could fall below 10.
- **The stimulus group was never used.** G1 (items 6–8) wasn't selected in any run. Its strict all-or-nothing rule would use 3 of 5 Strand 1 slots, and item 6 is very hard ($b = 2.96$).
- **The bank file contains a group header row** (`G1,GROUP,…`), and the blueprint's `model_type` dimension matches no bank column. PACER handled both correctly, deriving model type from the `model` column, but other tools may not.
- **The forms aren't strictly parallel.** The TIFs nearly match, but the TCCs differ by up to 0.67 raw-score points. Plan to equate through the 14 common items.

6 Sensitivity checks

To measure what drives the solution, I ran two variants with the same bank, TIF specification, and settings otherwise.

Scenario	Status	Time	TIF at θ = 0 (min of forms)	TIF at θ = −2 (min)	Shared per pair	On all 3 forms	Distinct items	p-value sums
As requested (overlap 10–15, p-value 12–14)	Optimal	44 s	5.88	3.02	15 / 15 / 15	14	29	12.04–12.14
Minimum overlap (exactly 10 per pair)	Feasible*	50 s	5.40	3.01	10 / 10 / 10	7	37	12.04–12.06
No p-value constraint (overlap 10–15)	Optimal	5 s	6.28	3.00	15 / 15 / 14	12	28	9.42–10.00

*Stopped at a 50-second limit with a valid but not proven-optimal solution. Every scenario met all its constraints. Shared per pair is listed as A–B / A–C / B–C.

- **The p-value rule costs about 6% of center information** (5.88 vs. 6.28) and makes the problem roughly 9 times slower to solve. Given concern 1, it is currently paying that price to control an unreliable number.
- **Minimum overlap costs about 8% of center information** (5.88 vs. 5.40) but halves the core shared by all forms and spreads the polytomous enemies (4, 5, 99) across forms.
- **The θ = −2 floor is binding in every scenario** (3.00–3.02), which confirms it is a bank-capacity limit, not a side effect of the other constraints.

7 Recommendations

1. **Reconcile the p-value column with the IRT calibration** before using the p-value rule. If the IRT parameters are current, replace the p-value sum with a TCC or expected-score target. That would control difficulty in a way that matches the parameters the TIF is built on.
2. **Decide how much overlap you want, not just the allowed range.** If exposure matters, set the overlap closer to 10 (the minimum-overlap build is a valid alternative). If linking strength matters more, the current 14-item core is generous. For equating, about 20–25% of the test is a common rule of thumb, which is 4–5 items here.
3. **Add polytomous items, or relax the 5-per-form requirement.** With only 8 GPCM items, three of them mutual enemies, every form will share the same polytomous content.
4. **Write or recalibrate items for the tails**, especially below θ = −1, if the θ = −2 target of 3.0 is a firm requirement. Otherwise consider lowering it to match the bank.

5. **Equate the forms** using the common items. The TCCs differ enough (up to 0.67 points) that raw scores aren't interchangeable.

Method notes

- **Assembly was done in PACER 3.2.0** (`assemble_test`, CP-SAT): three forms, simultaneous assembly, linking on, 10–15 common items, strict stimulus groups, D = 1.7, 180-second time limit. The constraint audit, solver status, objective, and form curves are PACER's output. The sensitivity runs used the same settings with a 50-second limit. The no-p-value scenario used a copy of the blueprint with the p-value row removed; the original file was not changed.
- **Computed by me from PACER output and the bank file:** the item table, overlap counts, exact TIF values at the spec points (matching PACER's curves to four decimals), TCC differences, the IRT-implied proportion correct (3PL, $\theta \sim N(0,1)$, D = 1.7), the information bounds in concern 4, and p-value sums. PACER's audit reports p-value sums on a ×10,000 scale (e.g., 121389 = 12.14).
- No identifying information about the requester is included in this report.